Machine Learning - Undergraduate Level Assessment 19 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following can only be used when training data are linearly separable?
   (a) Linear hard-margin SVM.
   (b) Linear Logistic Regression.
   (c) Linear Soft margin SVM.
   (d) The centroid method.

2. Statement 1| L_2 regularization of linear models tends to make models more sparse than L_1 regularization. Statement 2| Residual connections can be found in ResNets and Transformers.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

3. What is the dimensionality of the null space of the following matrix? A = [[1, 1, 1], [1, 1, 1], [1, 1, 1]]
   (a) 0
   (b) 1
   (c) 2
   (d) 3

4. MLE estimates are often undesirable because
   (a) they are biased
   (b) they have high variance
   (c) they are not consistent estimators
   (d) None of the above

5. Which image data augmentation is most common for natural images?
   (a) random crop and horizontal flip
   (b) random crop and vertical flip
   (c) posterization
   (d) dithering

True / False (10 marks)

Answer True or False

6. True or False: Adding more basis functions in a linear model decreases variance, leading to a more generalized model performance.
7. True or False: ImageNet contains images of various resolutions, providing a diverse dataset for training machine learning models.
8. True or False: Discriminative approaches model the conditional probability p(y|x, w) to predict outcomes based on input features and parameters.
9. True or False: Principal Component Analysis (PCA) is an unsupervised learning technique used for dimensionality reduction.
10. True or False: Overfitting is more likely when the hypothesis space is small, as simpler models tend to capture all patterns.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the n'th perrin number using recursion.
12. Write a python function to get the first element of each sublist.

End of Paper